

Spencer Perkins

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Skills

Languages: Scala, Java, TypeScript, Python, C, C++, Bash, Cypher, SQL, HTML, CSS, Dhall

Frameworks / Tools: Git, Angular, React, PyTorch, Playwright, Cypress, SBT, Maven, Gradle, Conda

Products: GitHub, Neo4j, PostgreSQL, MongoDB, Redis, Auth0, RabbitMQ, SendGrid, Jira, Confluence

DevOps & Cloud: Kubernetes, Docker, GitHub Actions, Argo CD, Google Cloud, AWS, Grafana, Jenkins, Prometheus, Helm, Terraform, Self-hosting

Experience

Full-Stack Software Engineer — Garner Distributed Workflow — Full-time

05/2024 – 09/2025

- Led an end-to-end **production database migration** containing sensitive data for **Fortune 10** clients (over 1 billion nodes) to ONgDB (open-source graph database), owning **Kubernetes/cloud infrastructure**, open-source code maintenance, benchmarking, and query optimization.
- Designed and implemented a domain-specific language (DSL) for logistical filtering, including in-memory evaluation, a strongly typed **GraphQL API**, and a migration layer to and from the legacy filtering system.
- Built fault-tolerant **Scala ZIO** data ingestion pipelines and scheduled **cron jobs** to reliably process millions of rows of financial and logistical data daily through **RabbitMQ** queues into **Neo4j**.
- Delivered features, integrations, and **UI/UX** for **self-signup**, filtering, and reporting in an **Angular** frontend requiring cross-team collaboration with design and product teams.

Machine Learning Researcher — Pardee Lab — Full / Part-time

05/2023 – 09/2025

- Implemented a **distributed deep learning** training platform on **Google Cloud** using **MongoDB** to run a PyGAD genetic algorithm with a **PyTorch** VAE + LLM fitness predictor, reducing runtime by $\sim 20\times$.
- Conducted interdisciplinary research in biology and machine learning, leading to accepted papers at **ICLR** and **NeurIPS** (<https://openreview.net/forum?id=sfTXIiu7Op>, <https://openreview.net/forum?id=DmX8LSSubU>).

Projects

Recipe Website — github.com/SpecialThing44/recipe-website — cooking.spencers.cc

2025

- Designed my own Recipe website using a Neo4j-backed social data model with strong typing to support **analytical queries**, personalized recommendations, and complex filtering over large graph datasets.
- Implemented a **RESTful API** using **Scala Play** and ZIO, emphasizing functional design and type safety.
- Fully **self-hosted** on Linux using **Docker**, with authentication, email delivery, object storage, and production deployment automation.

KV Store — University of Toronto

2025

- Designed and implemented core components of a **C++ LSM tree** backed key-value store with Bloom Filters, custom buffer management with LRU eviction, and performance benchmarking for latency and throughput.

Interactive Research Platform — University of Toronto

2023

- Worked in an Agile team to develop a **React/Java/MySQL** platform enabling real-time interaction between humans and software agents in decision games using a microservice architecture.
- Designed a modular API for website-game interaction and a **Python** framework for creating games.

Java Tamagotchi Game — University of Toronto

2022

- Built a Tamagotchi-style game in **Java** with a Swing UI, following **MVC** and **SOLID** principles, including a database for online user interaction and saved games.

Education

University of Toronto — Toronto, Ontario

Expected 05/2026

- Bachelor of Science — Computer Science Specialization — Arts & Science Internship Program
- **Relevant Coursework:** Operating Systems, Algorithms, Machine Learning, Software Engineering, Networking, Computer Security, Web Programming, Software Testing, Databases, Data Structures